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Presentation Letter

I am an AI Engineer and solution architect who designs, builds, and ships production LLM systems end to end, from shaping the solution with clients in pre-sales through architecture, implementation, cloud deployment, and knowledge transfer. My focus is the layers that decide whether an LLM system works in production: evaluation, retrieval and data design, and compliance. I build evaluation harnesses that show where accuracy is lost, use them to drive architecture and roadmap decisions, and benchmark model, cost, and latency trade-offs up front.

I have worked remotely with US companies delivering agentic AI to clients in healthcare, travel, media, and real estate: HIPAA-compliant document pipelines, multimodal document automation integrated with ERPs, NL2SQL and RAG assistants, and real-time voice agents that handle calls and operate web applications. Before that, I spent years in industrial AI in Brazil, taking ML from prototype to production: predictive maintenance and MLOps on AWS for wind turbines, and edge ML, computer vision, and IoT products.

I am looking to lead AI engineering and system architecture: setting technical direction, making trade-offs explicit, working closely with clients and stakeholders, and raising the engineering bar of the teams I work with.

Professional Experience

Zazmic Inc — United States of America (Remote)

AI/ML Engineer — *Dec. 2025 – Present*

- **Solution Architecture & Technical Ownership**: Technical owner of generative-AI engagements on GCP across healthcare, travel, and media, from pre-sales demos and estimates through architecture, delivery, and handover.
- **Evaluation-Driven Engineering**: Built evaluation harnesses, LLM-as-Judge suites, and trace-based root-cause analysis that pinpoint where accuracy is lost and steer architecture and roadmap decisions.
- **Retrieval & Data Architecture for Agents**: Designed data models, retrieval strategies, and NL2SQL agents with abstention policies, and built RAG pipelines with incremental ingestion and citation-grounded answers.
- **Regulated (HIPAA) AI Systems**: Architected event-driven document pipelines processing PHI (Document AI, Gemini, Pub/Sub, Cloud Run) with compliance by design, architecture decision records, and full monitoring.
- **Real-Time Multimodal Agents**: Shipped voice agents that operate web applications through live native-audio models and function calling (Gemini Live, ADK, FastAPI WebSockets, React).
- **Multi-Agent Pre-sales Automation**: Built a LangGraph multi-agent tool with human-in-the-loop review that turns discovery calls into cloud proposals, backlogs, and SOWs.

Dry Ground AI United States of America (Remote)

AI Engineer *Mar. 2025 – Nov. 2025*

- **AI Voice Agent Development**: Developed and deployed AI voice agents to answer real estate-related calls; integrated tools to retrieve up-to-date information, send messages and emails, and automate client interactions.
- **Client Profiling & Lead Generation**: Developed a data science workflow for classifying user profiles and identifying potential clients, supporting strategic sales initiatives.

WEG Jaragua do Sul, Brazil

Data Scientist *Jun. 2024 – Mar. 2025*

- **Wind Turbine Anomaly Detection**: Deployed a production anomaly detection system on AWS SageMaker anticipating failures up to two weeks ahead and flagging ∼70% of maintenance interventions; companion explainability API surfaced root-cause diagnostics for maintenance engineers.
- **End-to-End MLOps**: Built a reproducible MLOps pipeline (MLflow, Docker, Kubernetes, AWS) spanning automated training, versioning, deployment, and live monitoring via Grafana dashboards backed by InfluxDB, enabling iterative model improvements at production scale with full experiment lineage tracking.

Macnica DHW Florianópolis, Brazil

Data Scientist *Nov. 2022 – Oct. 2023*

- **Smart Sensors**: Developed an SVM-based classification pipeline for vibration sensor data; applied multi-objective Genetic Algorithms (NSGA-II) to optimize the trade-off between model accuracy and computational efficiency for edge hardware deployment.
- **Computer Vision APIs**: Built a dual-layer video anonymization API using OpenCV and DBSCAN clustering for motion-based person detection, with a pre-trained ResNet-50-FPN model (PyTorch) as a verification layer; optimized specifically for low frame-rate surveillance footage.
- **IoT Development**: Led end-to-end IoT product development with full Docker containerization of the edge-to-cloud stack (MQTT Broker, PostgreSQL, InfluxDB, Grafana, Python services); implemented edge computing including FFT calculations directly on Senspider hardware.

Technical Expertise

Architecture & Practices:	Solution architecture, ADRs, event-driven design, HIPAA compliance.
Programming Languages:	Python, C, C++, Bash (Unix Shell).
LLM & Agentic Frameworks:	Google ADK, LangGraph, LangChain, Gemini Live, Vertex AI RAG, Claude.
Evaluation & Tracing:	LLM-as-Judge, Langfuse, LangSmith, OpenTelemetry, Grafana.
ML & Deep Learning:	PyTorch, Hugging Face, scikit-learn, XGBoost.
Cloud Platforms:	GCP (Vertex AI, Document AI, Cloud Run, Pub/Sub), AWS (SageMaker).
MLOps & Deployment:	Docker, Kubernetes, MLflow, Git, GitHub Actions CI/CD.
Full-Stack:	React, TypeScript, FastAPI, WebSockets, SSE streaming.
Data Engineering:	PostgreSQL, Cloud SQL, Firestore, BigQuery, Redshift, InfluxDB.
Embedded Systems:	ARM Cortex-M, Jetson Nano, Arduino, FreeRTOS, MQTT.
Optimization:	Bayesian Optimization, Multi-objective Evolutionary Algorithms.

Certifications

Claude Certified Architect – Foundations, Anthropic *Jul. 2026*

Research Experience

University of Warwick, Warwick Business School Coventry, United Kingdom

Research *Oct. 2019 – Apr. 2022*

- Bayesian optimization and multi-objective MCTS for supply chain scheduling

Shinshu University Nagano, Japan

MSc Research *Apr. 2017 – Mar. 2019*

- Multi-objective evolutionary algorithms; Best Student Paper Award, EMO 2019 (East Lansing, MI, USA)

Education

Shinshu University Nagano, Japan

Master of Engineering in Electronics and Information System *Apr. 2017 – Mar. 2019*

- MEXT Monbukagakusho Scholarship (Japanese Government)

Concordia University Montreal, Canada

Exchange Student, Electrical Engineering *Aug. 2012 – Aug. 2013*

- Science Without Borders Scholarship (Brazilian Government)

Federal University of Technology - Parana Curitiba, Brazil

Bachelor of Engineering in Electronics *Jan. 2010 – Jul. 2016*

Languages

Portuguese: Native

English: Fluent

Japanese: Conversational